DAVID MGHONGO MWANDAIRO

DATA ENGINEER

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PROFESSIONAL SUMMARY

Data Engineer with 7+ years of experience designing, developing, testing and maintaining batch, streaming and real-time data pipelines and curated analytical datasets, including in banking, financial services and other regulated data environments. Proven record delivering production ETL/ELT pipelines with Python, SQL, Apache Kafka, Apache Spark, Apache Airflow and dbt into Snowflake, BigQuery and ClickHouse, building reusable data products that power business intelligence, analytics and AI solutions, including the data infrastructure behind a proprietary credit-scoring and machine learning system. Experienced implementing data quality controls, reconciliation, metadata, lineage and monitoring (dbt tests, Great Expectations, Prometheus, Grafana), alongside RBAC, access controls, encryption, backup, disaster recovery and SOC2-aligned auditability. Skilled in automated testing, Git source control, Docker and CI/CD to support deployments across development, testing and production environments. Collaborates with cross-functional teams and business data owners to translate business requirements into fit-for-purpose data solutions, with a strong drive to extend into unstructured data pipelines for generative AI.

CORE COMPETENCIES & TECHNICAL SKILLS

Data Pipelines & Orchestration: Batch, Streaming & Real-Time Data Pipelines, ETL/ELT Development, Apache Kafka, Apache Spark (incl. Spark Streaming concepts), Apache Airflow, dbt (Data Build Tool), Change Data Capture (Debezium), Airbyte, Event-Driven Pipelines, REST API & Webhook Integration, Custom Python & SQL Connectors, Pipeline Performance, Reliability, Scalability & Cost Optimization.

Data Warehousing, Modeling & Data Products: Snowflake, BigQuery, ClickHouse (OLAP), MySQL & PostgreSQL (OLTP), Dimensional Modeling (Kimball, Star & Snowflake Schemas), Curated Analytical Datasets, Reusable Data Products & Data Marts, Feature Sets for ML & AI, Semantic Layer, Data Lake & Lakehouse Concepts, NoSQL Concepts.

Data Quality, Governance & Observability: Data Quality Controls, Reconciliation, Metadata Management, Data Lineage, Automated Data Validation (dbt Tests, Great Expectations), Anomaly Detection, Prometheus, Grafana Dashboards & Alerting, Monitoring, Logging & Metrics (Latency, Throughput, Failure Rates, Data Freshness), Incident Response & Troubleshooting.

Security, Privacy & Resilience: Data Security & Privacy, Role-Based Access Control (RBAC), Access Controls, Data Encryption (At Rest & In Transit), Data Retention, Auditability, High Availability, Disaster Recovery & Recovery Planning, Automated Database Backups, Regulatory Compliance (SOC2), Operational Resilience.

DevOps & Engineering Practices: CI/CD Pipeline Automation, Automated Testing, Git Source Control, Docker Containerization, Linux Infrastructure, Deployments across Development, Testing & Production Environments, Technical Documentation & Operating Procedures.

Programming & Scripting: Python (Pandas, NumPy, Modular Pipeline Automation), SQL (Complex Querying & Index Tuning), Bash / Shell Scripting, R for Statistical Modeling.

Collaboration & Leadership: Cross-Functional Collaboration with Data Owners, Data Engineers, Analysts & Senior Stakeholders, Translating Business Requirements into Datasets, Mentoring Junior Engineers, Remote Team Collaboration, Extreme Ownership.

PROFESSIONAL EXPERIENCE

Data Engineer | **Data Science East Africa** | *Nairobi, Kenya* | *May 2026 to Present* | *Hybrid, Full-Time*

- Supported ETL workflows and data pipeline development, integrating multiple source systems for inventory and logistics analytics.
- Collaborated cross-functionally to optimize data pipelines, increasing operational efficiency by 15%.
- Performed data validation and implemented governance and access controls, increasing data integrity by 20% and supporting secure, reliable analytics pipelines.
- Used Grafana and Tableau for dashboarding and real-time analysis of streaming pipelines, supporting platform observability and monitoring.

Data Engineer | **Lipa Later** | *Nairobi, Kenya* | *Feb 2023 to Oct 2025* | *Hybrid, Full-Time*

- Architected and maintained scalable batch and real-time ETL/ELT pipelines ingesting and standardizing transaction data from 50,000+ merchant partners and 1 million+ active customers across four countries, processing $100M+ in annual credit volume while reducing pipeline latency by 40% to enable near-real-time decisioning.
- Designed and deployed a centralized data warehouse consolidating the core banking system, payment gateways and merchant APIs, reducing daily portfolio risk report generation time from 6 hours to 45 minutes.
- Built and optimized the data infrastructure and feature pipelines powering a proprietary credit-scoring and ML system, reducing approval turnaround from 48 hours to under 5 seconds.

- Developed real-time fraud detection pipelines monitoring behavioral metrics and transaction velocity, flagging 2,000+ suspicious transactions daily and improving detection accuracy by 30% with no impact to checkout experience.
- Engineered secure data integrations with open finance partners (Mono, Pngme) to enrich customer profiles, increasing approval rates by 25% for previously unbanked customers.

Data Engineer | **Bench Accounting** | *Remote* | *Feb 2020 to Dec 2022* | *Full-Time*

- Architected and maintained high-volume ETL/ELT pipelines (Python, Airflow, dbt) ingesting and standardizing financial transactions from 20,000+ SMBs across 50+ financial institutions, scaling to 30 million monthly transactions while reducing end-to-end pipeline latency by 35%.
- Developed automated data quality monitors and reconciliation protocols detecting 1,200+ daily data anomalies, improving bookkeeping data accuracy by 25% and saving the operations team 20 hours per week.
- Designed and deployed a centralized cloud data warehouse (Snowflake, BigQuery) consolidating 15 disparate source systems, reducing monthly financial report generation time from 3 days to 2 hours.
- Implemented a semantic data layer and curated data marts enabling 50+ non-technical stakeholders to self-serve queries, cutting ad-hoc engineering requests by 60%.
- Engineered secure, high-performance data export APIs and migration pipelines, cutting full client data migration time from 5 days to under 4 hours while ensuring SOC2 and financial audit compliance.

Data Engineer / Laboratory Data Analyst | **Burn Manufacturing** | *Ruiru, Kenya* | *Dec 2017 to Dec 2018*

- Developed modular Python and Bash/Shell scripts to extract, transform and load time-series telemetry and quality testing datasets from manufacturing IoT sensors and testing equipment, reducing manual data preparation time by 40%.
- Automated validation checks and standardized data recording structures, boosting dataset reliability and completeness by 25% across factory and field data collection.
- Engineered analytics-ready performance datasets across 50+ product models, applying statistical modeling and hypothesis testing (Python/R) to guide manufacturing decisions.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Analytical Chemistry with Management | Kenyatta University, Nairobi, Kenya

Applied Data Science Lab Certification | WorldQuant University

Certified Data Engineer | LuxDev Academy

LANGUAGES & KEY COMPETENCIES

Languages: English (Fluent), Kiswahili (Native), French (Conversational).

Key Competencies: Instills trust through transparent data lineage and honest communication; acts with courage in addressing infrastructure bottlenecks and taking full ownership of engineering workflows; makes informed decisions by seeking relevant telemetry and data input and escalating issues effectively; thrives in collaborative, high-impact environments.